CIF425 — Introduction To Large Language Models

Term Project

2025–2026 Spring Semester

Field	Details
Submission Deadline	May 13, 2025 — 23:59
Group Size	Individual or groups of 2
Group Registration	Notify the instructor via e-mail by April 10, 2025
Assessment	Code + Live Demo + Q&A

Overview

You are required to complete one of the four project options listed below. Your project does not need to achieve perfect results. What matters is that you have genuinely worked on it and that you can give consistent, informed answers to questions about your model, dataset, and code during the demo.

Option 1 — Encoder-Based Turkish News Application

Using a pretrained Turkish encoder model (BERTurk or (or other pretrained model you may use), you will apply fine-tuning to perform two separate tasks on Turkish news texts:

- Topic classification — categorizing a news article by subject
- Sentiment analysis — determining whether the content is positive, negative, or neutral

Technical Requirements
• BERTurk must be fine-tuned separately for each task
• A separate labeled dataset must be used for each task
• Datasets can be obtained from Kaggle, HuggingFace, GitHub, or similar platforms
• The application must run through a user interface (web, mobile, Gradio, Streamlit, etc.)

Option 2 — Decoder-Based Turkish Chatbot

Using a pretrained Turkish decoder model (Turkish GPT-2), you will apply instruction fine-tuning with a domain-specific Turkish dialogue dataset to build a chatbot that responds to questions within a chosen topic area.

Technical Requirements
• Turkish GPT-2 (or other pretrained model you may use) must be fine-tuned using a domain-specific Q&A or dialogue dataset
• The dataset can be obtained from Kaggle/HuggingFace or created manually
• Generating synthetic Q&A pairs with AI tools (ChatGPT, Claude, etc.) is permitted
• The chatbot must run through a user interface (web, mobile, Gradio, Streamlit, etc.)

Option 3 — Training a Mini Language Model from Scratch

You may build a small-scale Turkish language model from scratch. High performance is not expected. The focus is on going through the full training pipeline and documenting the process.

What to Report
• Architecture choice — encoder or decoder, and the reasoning behind it
• Dataset — source, preparation process, and size
• Training objective — what the model is trained to do
• Training curves — loss and validation loss over epochs

Note: A small-scale model (e.g., 2–4 Transformer layers, around 10M parameters) is sufficient. Make sure your training setup can run on Google Colab.

Option 4 — Custom LLM Project

You may propose and develop an original project related to large language models that covers topics discussed in this course.

Conditions
• You must contact the instructor by e-mail and receive approval before starting
• The project must be clearly connected to course topics (Transformer, fine-tuning, encoder/decoder, etc.)

Assessment

Projects will be evaluated through a live demo. During the demo, questions will be asked in the following areas:

- Model — architecture, why it was chosen, how it works
- Dataset — where it was obtained, how it was prepared, its size and structure
- Code — fine-tuning steps, key parameters, and any challenges encountered

Important: Perfect results are not expected. It is sufficient to demonstrate that you have genuinely worked on the project and that you understand what you have done.

Groups & Contact

- Projects may be completed individually or in groups of two
- Groups of two must notify the instructor via e-mail by April 10, 2025
- Students choosing Option 4 must contact the instructor for approval before starting
- Anyone with questions or who needs further clarification is welcome to get in touch